

3D modelling of masonry tower soil-structure interaction in OpenSees using mixed implicit-explicit material integration

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Towers in 2012 Emilia & 2016 Amatrice earthquakes



Amatrice's main street
Lazio, Italy

<https://www.theguardian.com/world/2016/aug/24/before-and-after-pictures-of-italian-towns-devastated-by-deadly-earthquake>

The inherent brittleness of masonry renders tower structures particularly dangerous.



The cathedral of Mirandola
Emilia-Romagna, Italy

<https://ilfattostorico.com/2012/06/05/i-danni-del-sisma-ai-beni-culturali-in-emilia/>

The research question

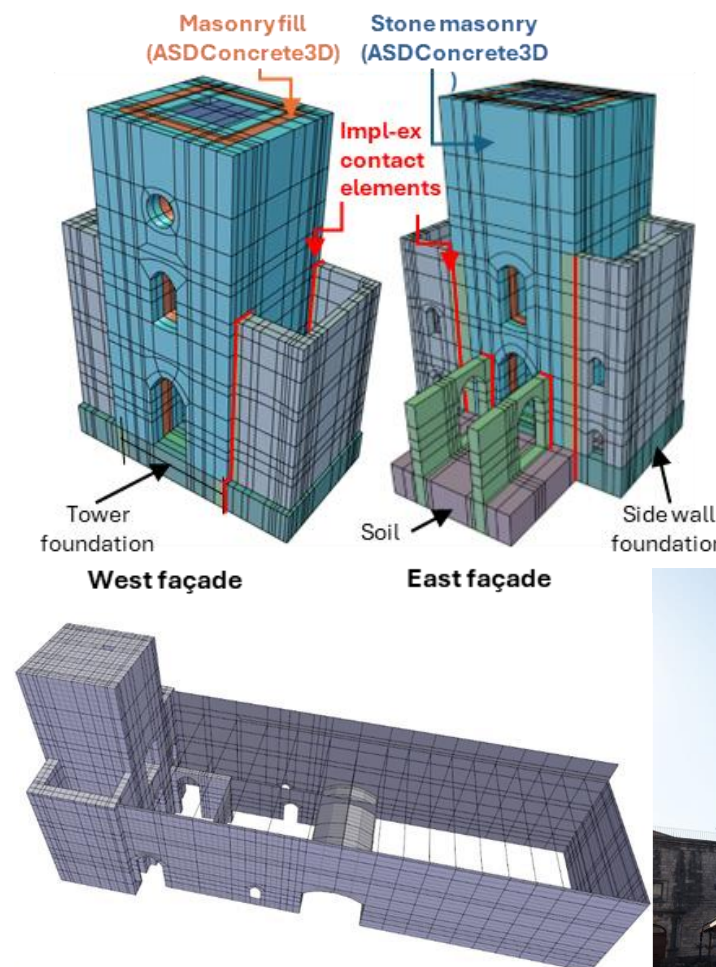
- Shallow-founded RC frame structure behavior has been extensively studied by
 - Bullock et al. (2019),
 - Macedo & Bray (2018),
 - Gazetas (2015),
 - Gelagoti et al. (2012) and many others.

How do the **stiffness degradation** and **strength** of the soil around the foundation affect the **cracking damage** predicted in masonry towers?

Gazetas (2015) discusses the beneficial effects of rocking isolation in RC wall-frame structures on footings during seismic shaking.

How does this apply to the bell tower?

The bell tower of Guardiafrede in Italy



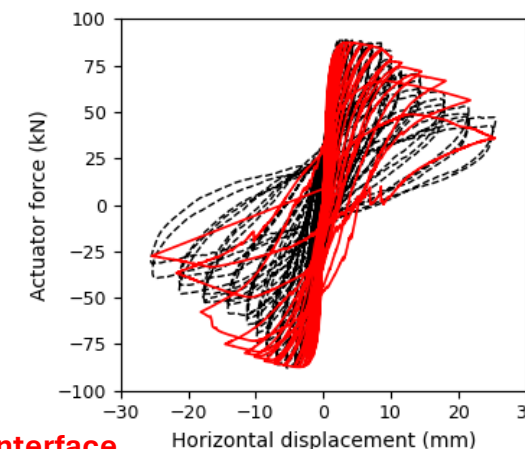
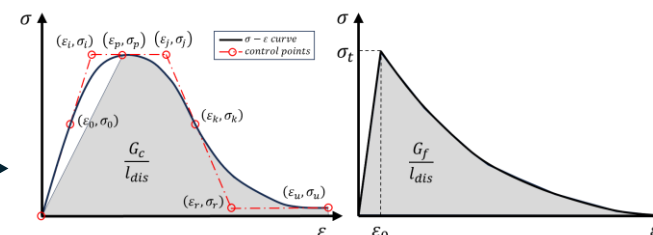
Material integration

Challenges

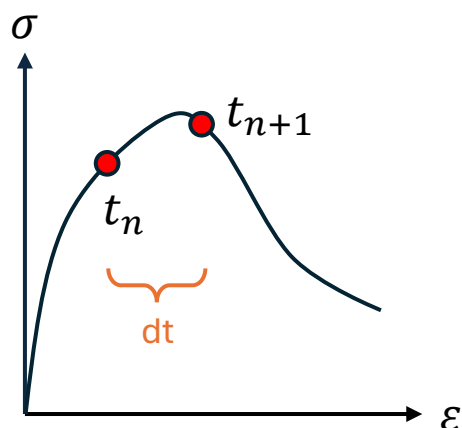
1. Masonry behaviour involves softening.
2. Dynamic contact-friction condition.

Solutions with traditional schemes are costly in large models!

ASDConcrete3D - Petracca et al. (2022)

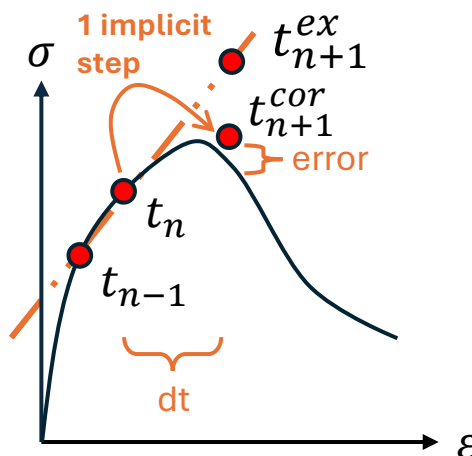


Implicit



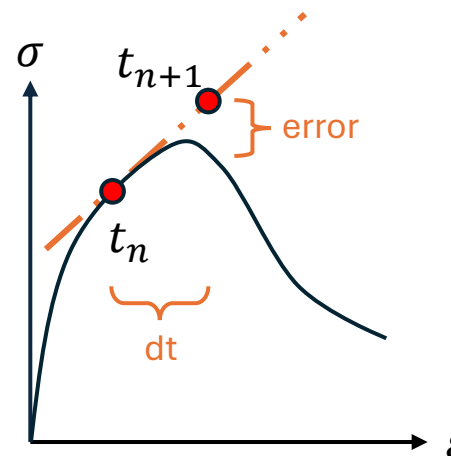
May not find solution
after 10+ iterations

Implicit-Explicit (IMPL-EX)



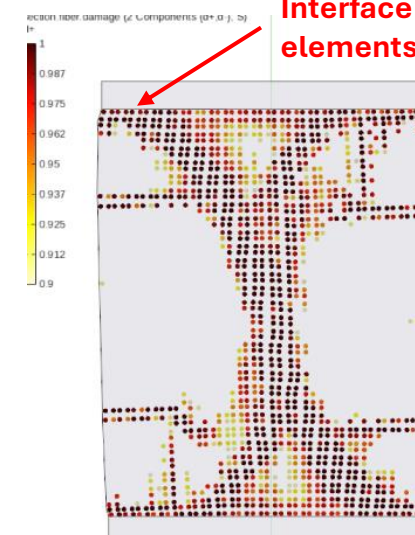
Error control
required!

Explicit

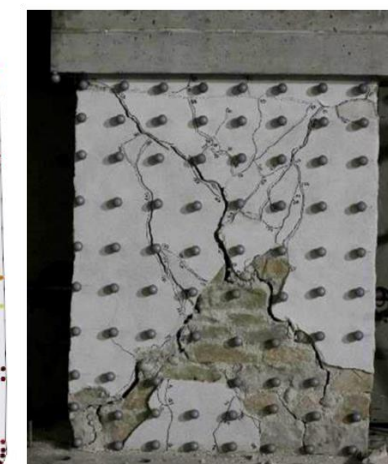


Time step may become
extremely small

video



ASDConcrete3D
calibration

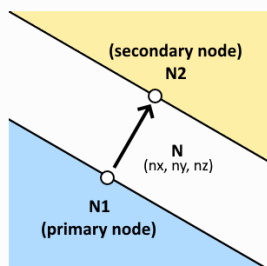


Specimen CT01
tested by
Senaldi et al. (2018)

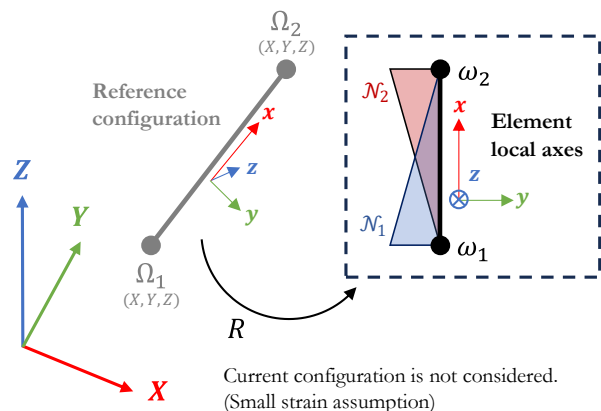
New implicit-explicit contact element in OpenSees

<https://opensees.github.io/OpenSeesDocumentation/user/manual/model/elements/zeroLengthContactASDimplex.html>

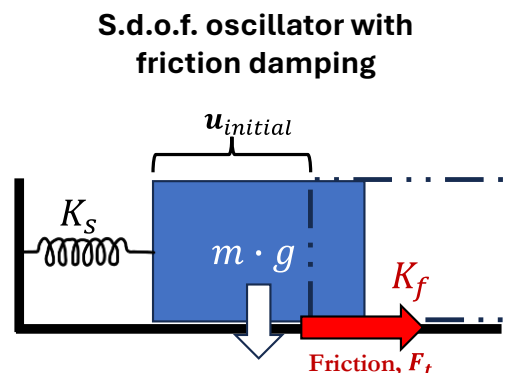
3.1.3.4. ZeroLengthContactASDimplex Element



- Penalty-based frictional MC contact element
- The two nodes can have different dofs Solids (3 DOFs) to Beams/Shells (6 DOFs) or UP-Solids (4 DOFs)

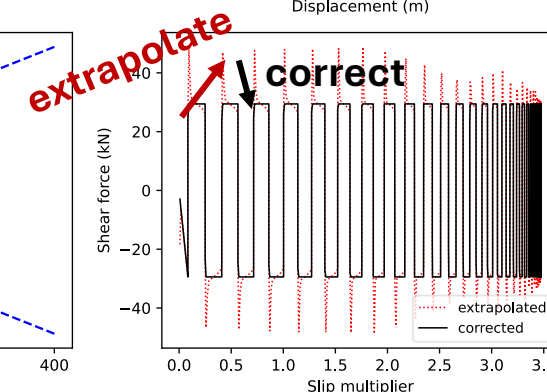
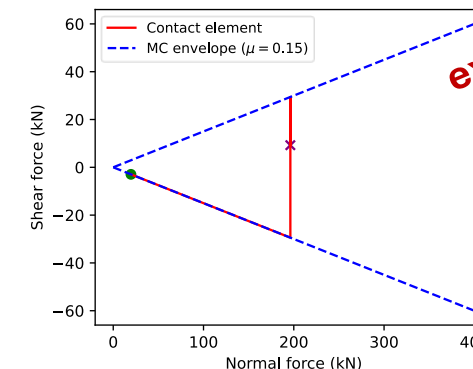
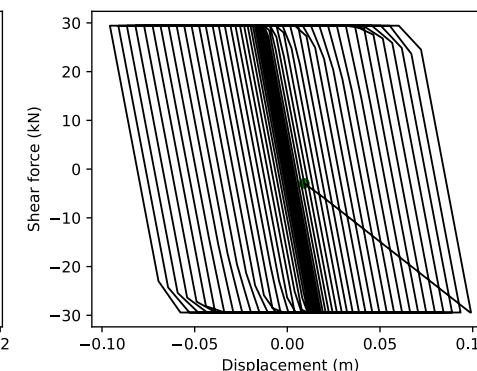
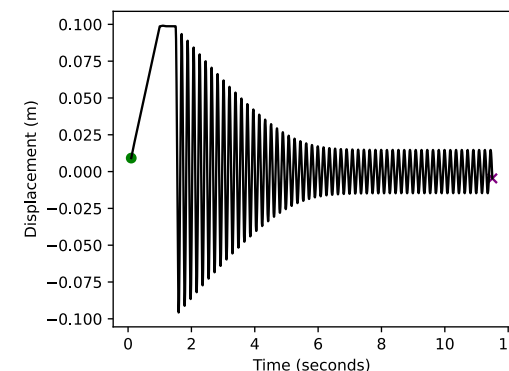


Reference configuration. The global coordinate system versus the contact element local axes.



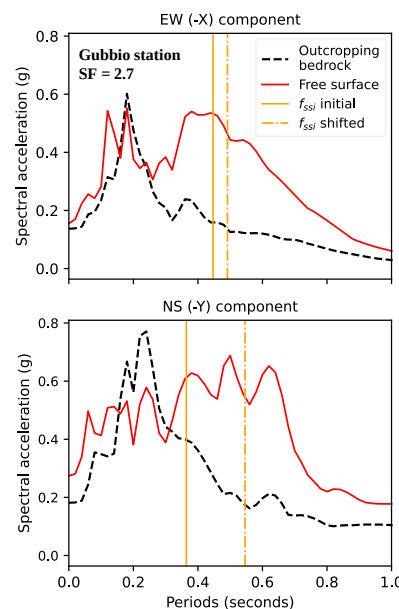
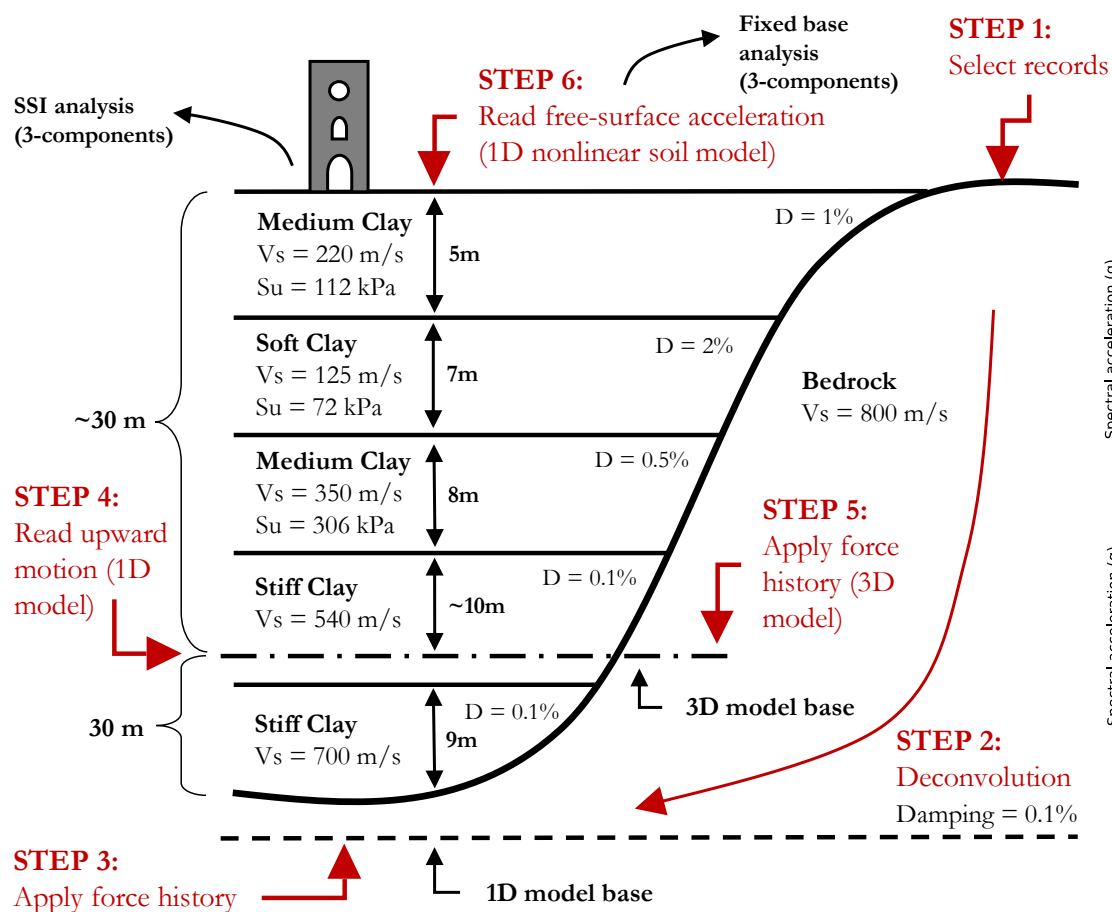
ALGORITHM

1. Explicit phase
Extrapolate internal variables
2. Newton-Raphson iteration
3. Implicit phase
Correct internal variables

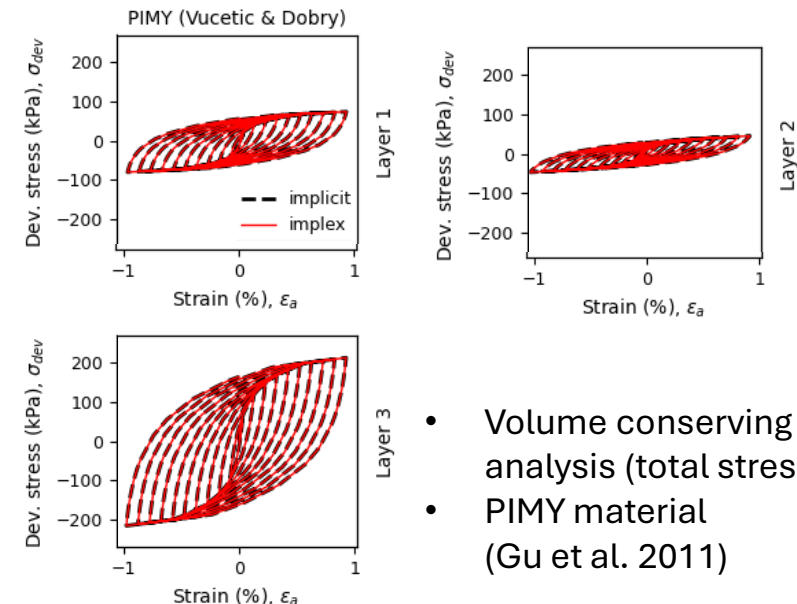


Soft impl-ex friction damper ($K_f = 2e3$ kPa). The discrepancy between the extrapolated and corrected stresses. (dt = 0.005 seconds)

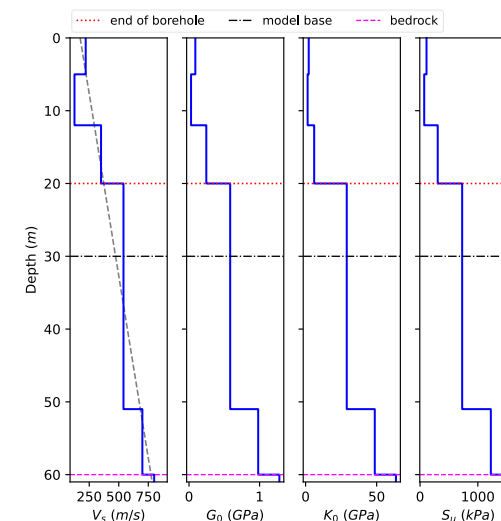
Methodology



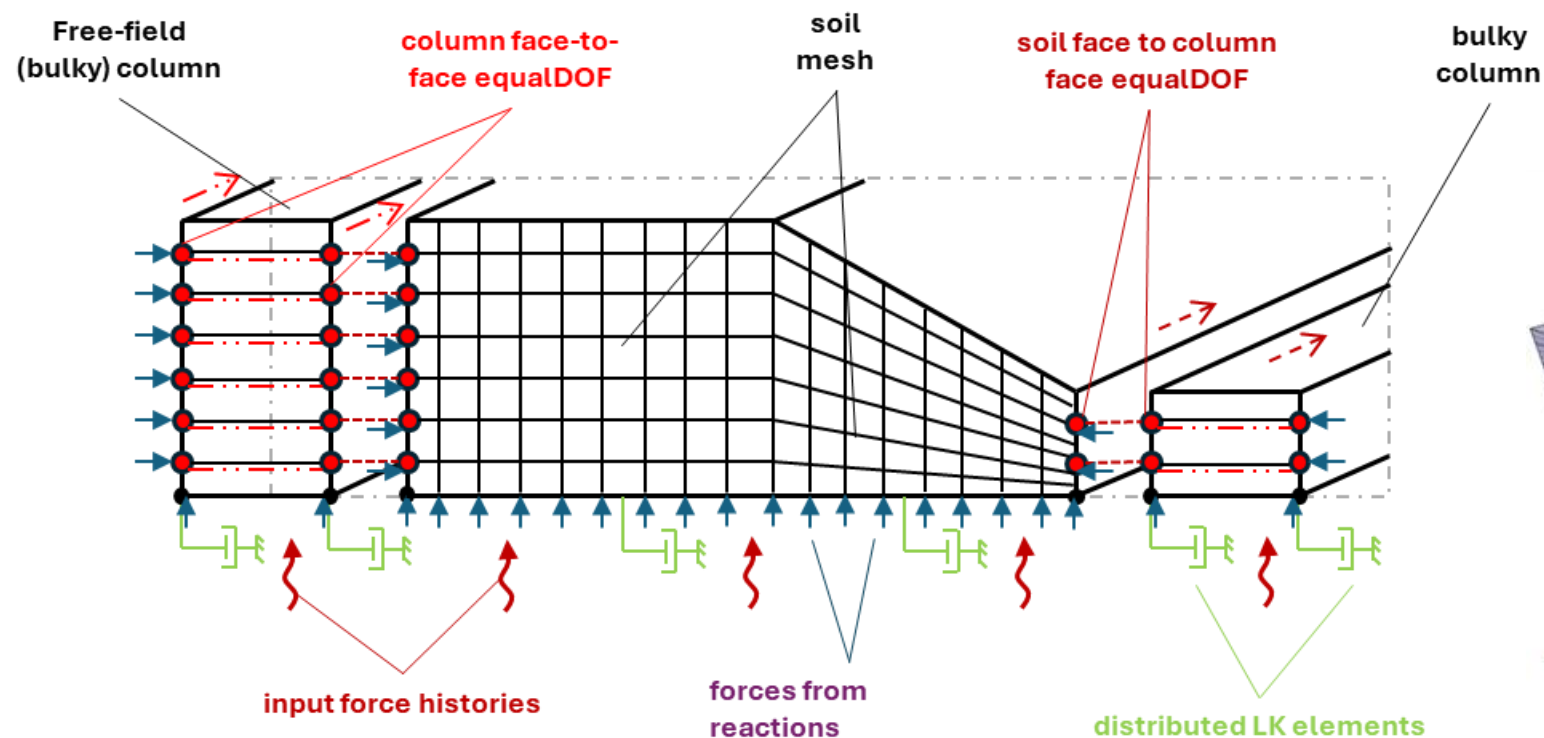
Umbria 1984, Mw 5.6



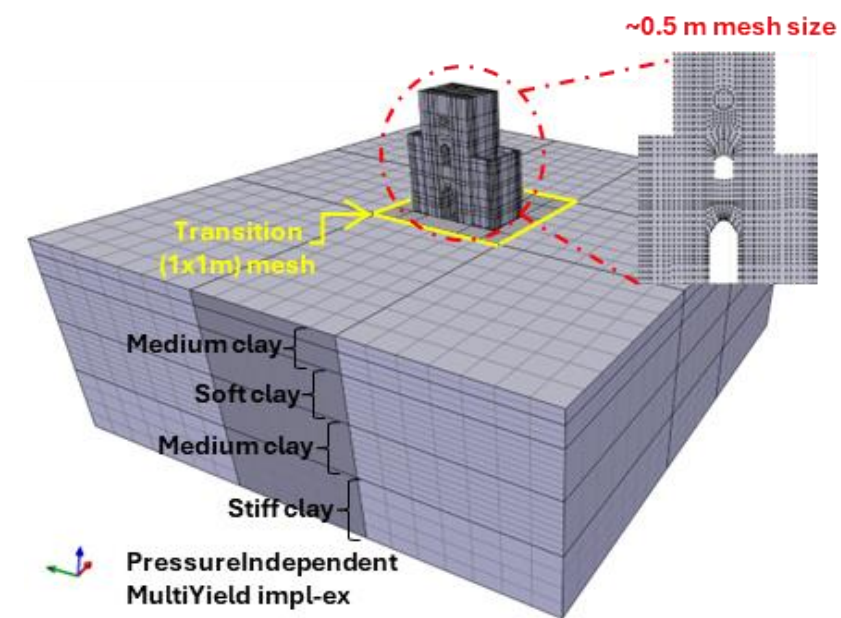
- Volume conserving analysis (total stress)
- PIMY material (Gu et al. 2011)



Boundary conditions



- 3D dissipating boundary conditions
- Extension of the free field bulky column approach used by Su et al. (2019)



Force boundary conditions applied to limit boundary displacements

Parallel solution

Analysis Parameters:

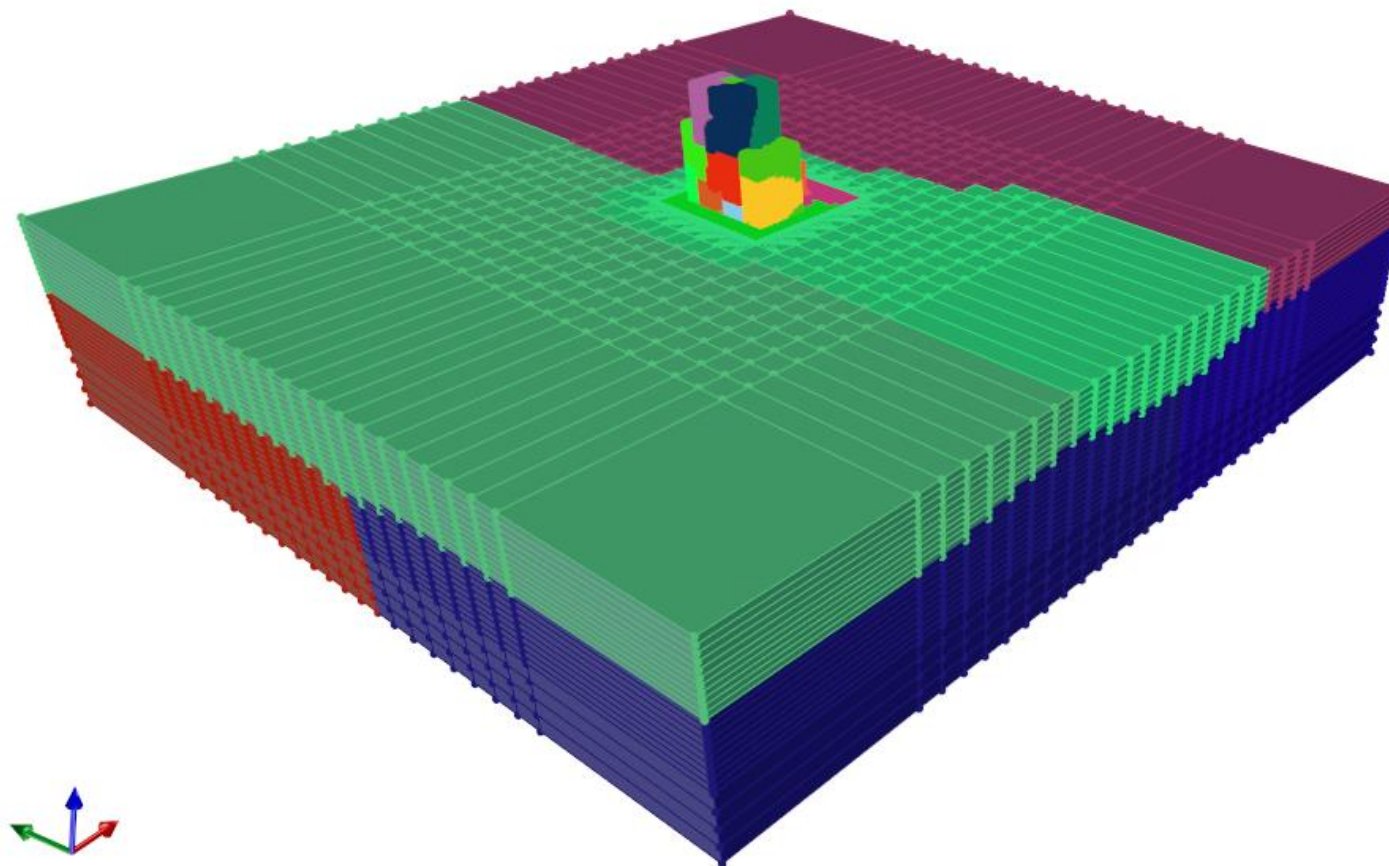
- $dt = 0.0006$ seconds
- TRBDF solver
- Penalty constraints: $1e12$
- MUMPS parallel matrix solver
- Krylov-Newton algorithm

32 Partitions to run in OpenSeesMP

~7,000 nodes & 10,000 elements per thread

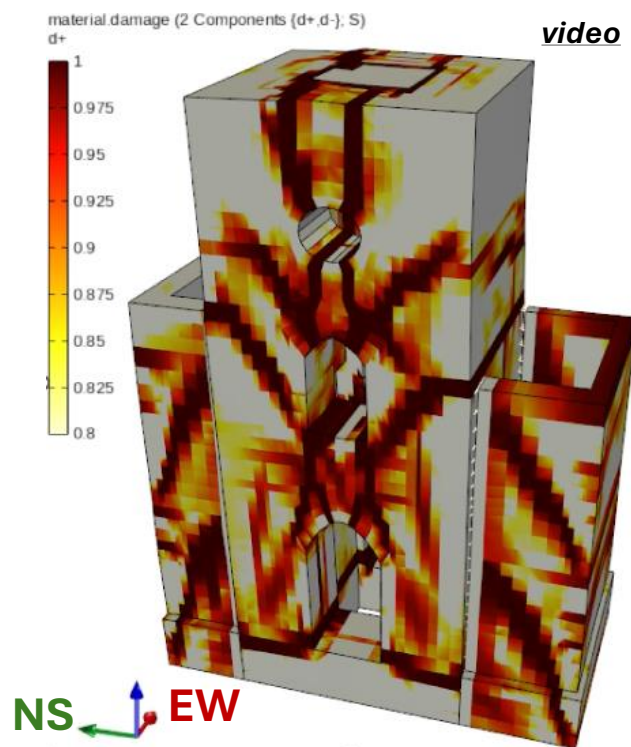
In total: 102465 nodes, 307,395
d.o.f., 219798 elements

9 seconds record takes **21 hours** in a i9-
13th gen. 32-core workstation



The IMPL-EX step size

Fixed base model without soil



Umbria 1984 earthquake

- Free surface acceleration
- 3-component motion

Tensile Cracking Damage

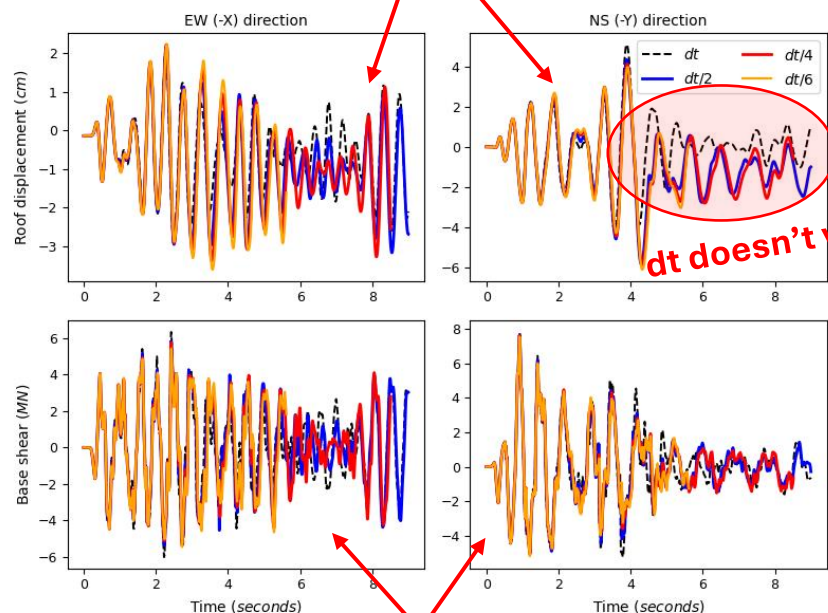
0: No damage

1: Full damage

Roof Displacements

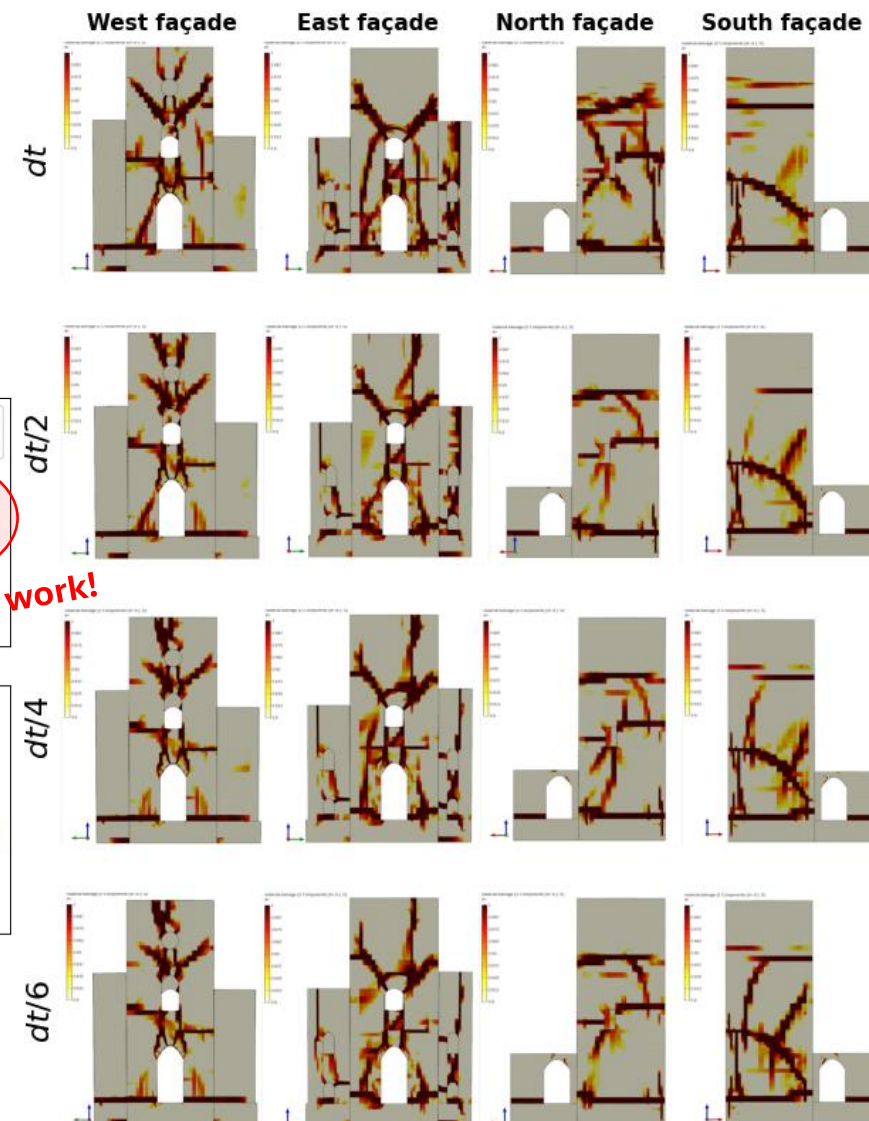
EW

NS



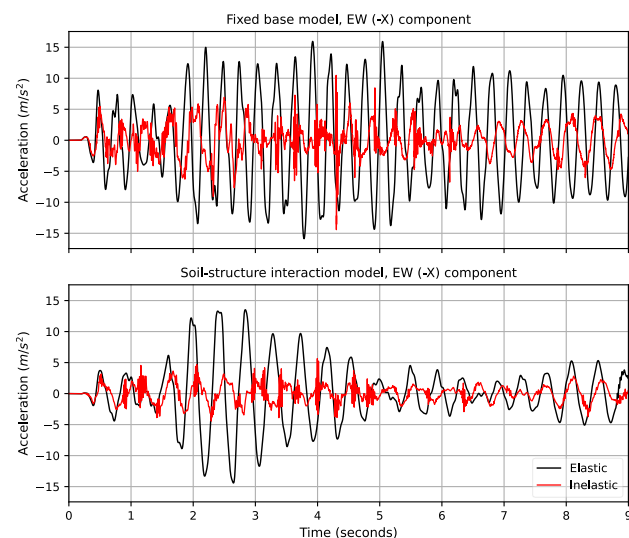
Base shear (MN)

*dt is the record sampling interval



Foundation rocking mechanism

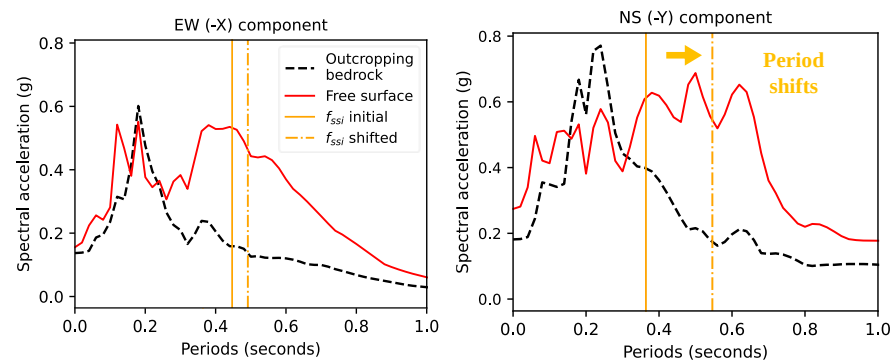
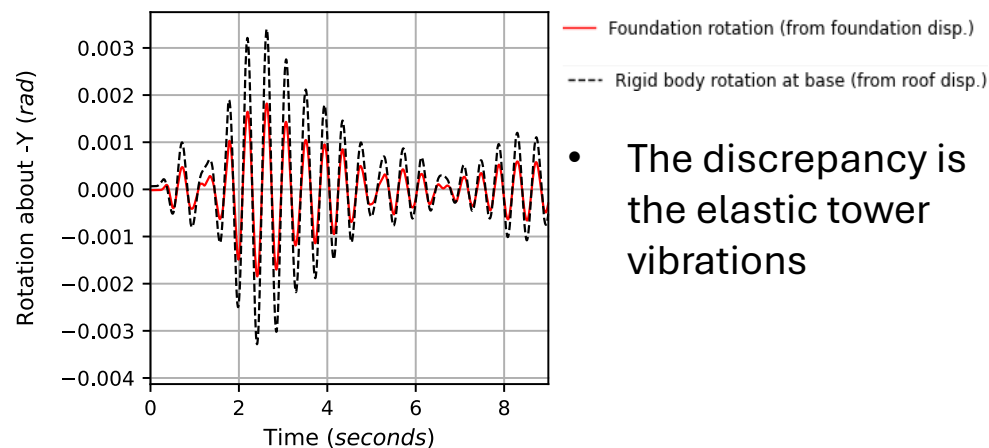
FIXED BASE NO SOIL



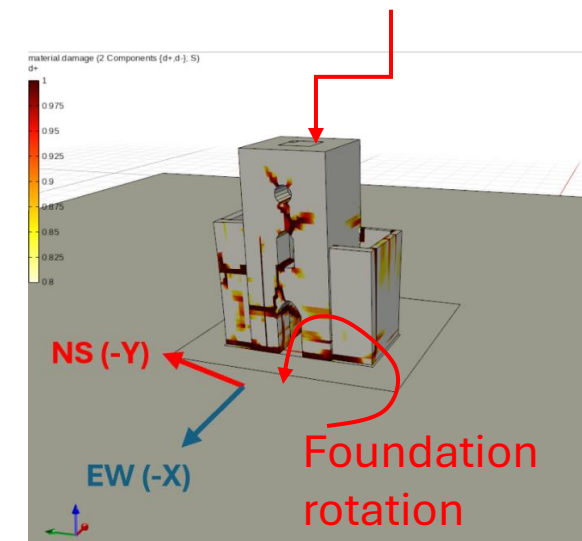
SSI MODEL

- Lateral response is governed by the foundation rotation
- A period shift is observed in the structure

Linear SSI model displacements EW component



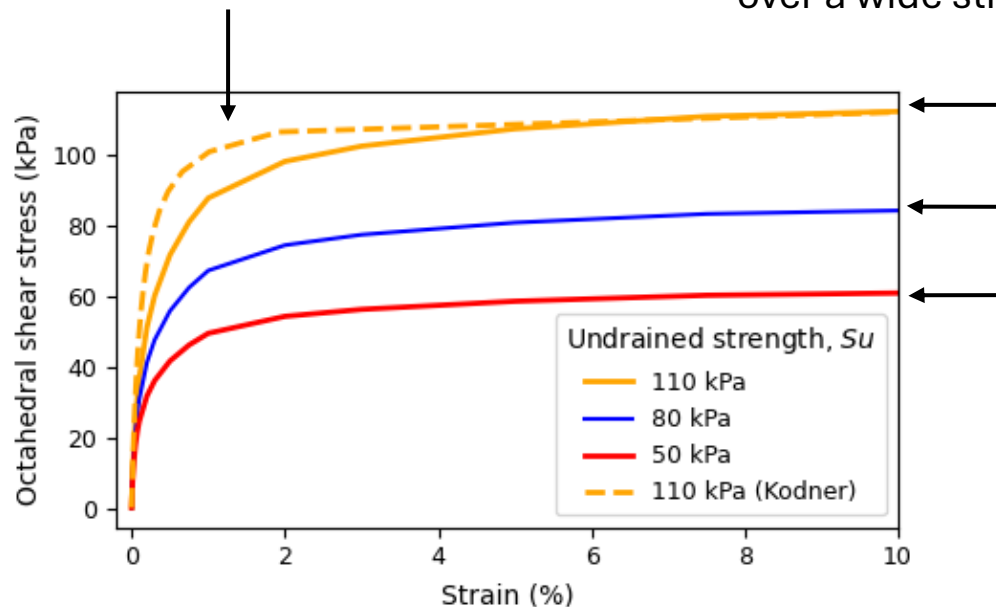
$$\text{Roof Drift} = \text{Disp.} / \text{Height}$$



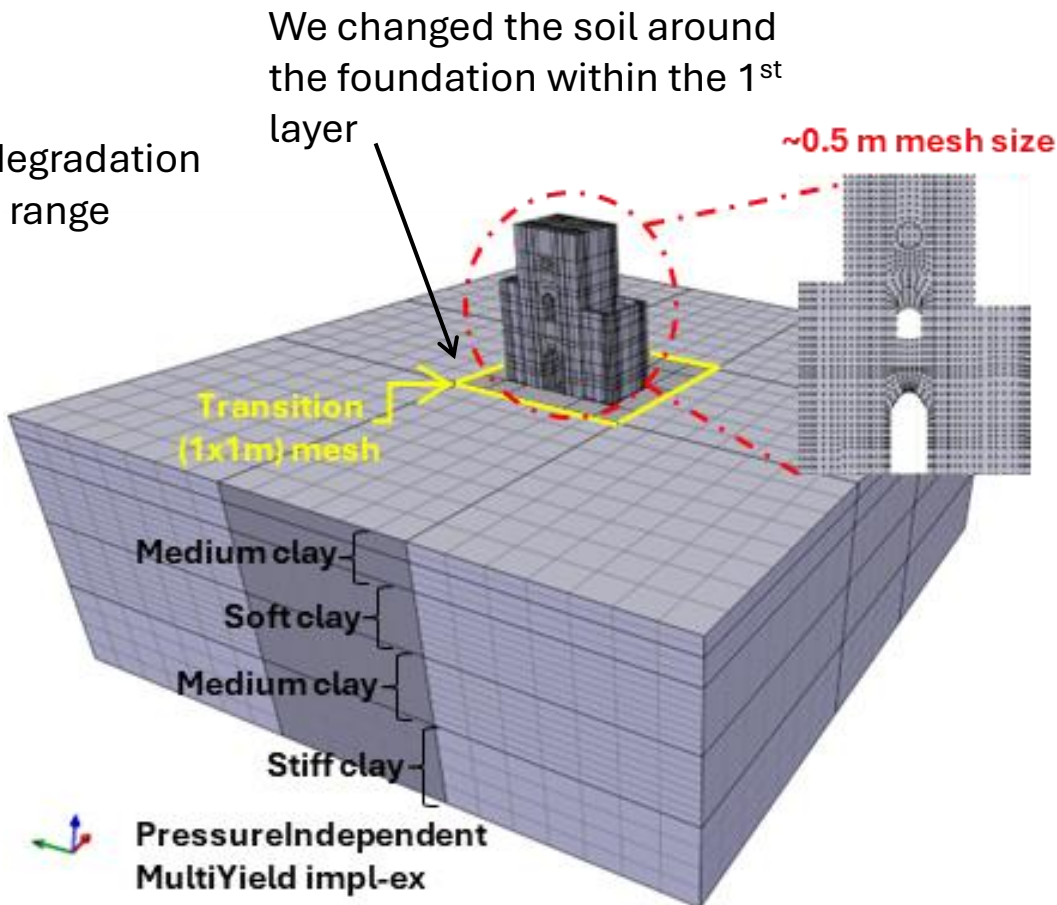
Rocking isolation mechanism

Faster stiffness degradation over a narrow strain range

Slower stiffness degradation over a wide strain range



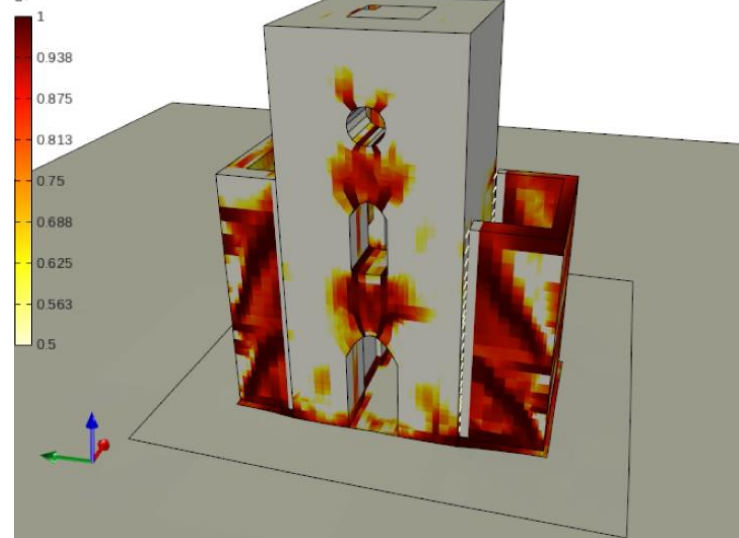
1. The effect of G/G_{max} characteristics
2. The effect of undrained strength



Rocking isolation mechanism

video

material damage (2 Components {d+, d-}; S)
d+

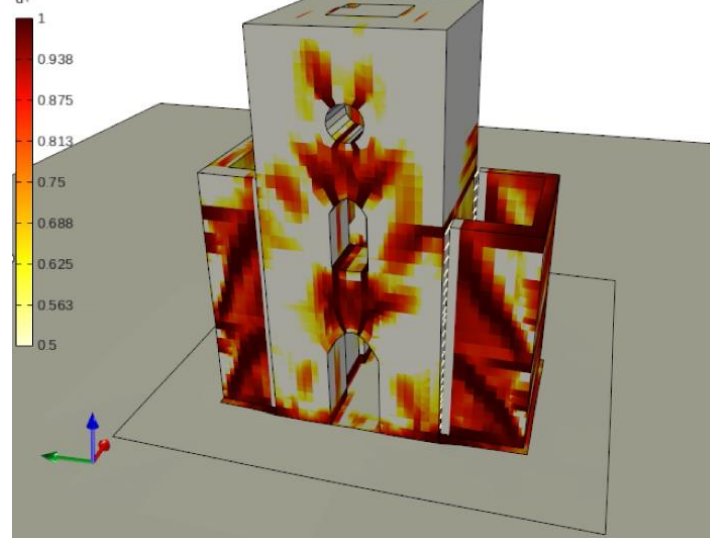


$S_u = 110$ kPa
(Vucetic & Dobry)

- Damage accumulation mostly during small soil strains

video

material damage (2 Components {d+, d-}; S)
d+

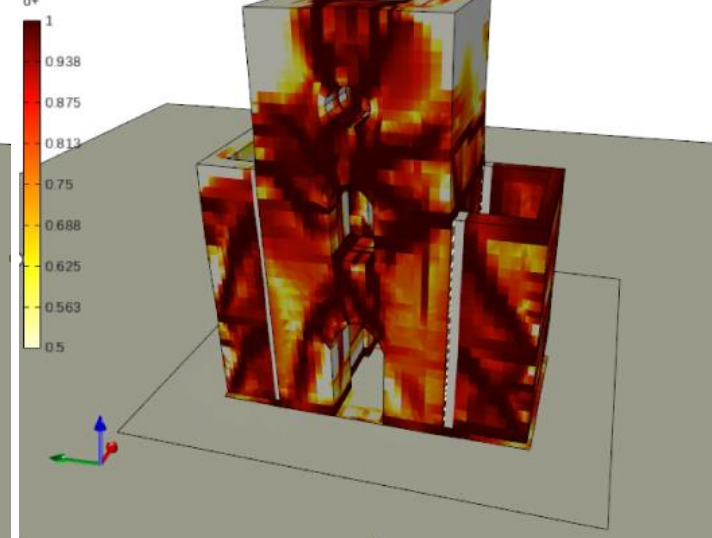


$S_u = 110$ kPa
(Kodner or Hyperbolic)

- Damage accumulation continues after the peak acceleration

video

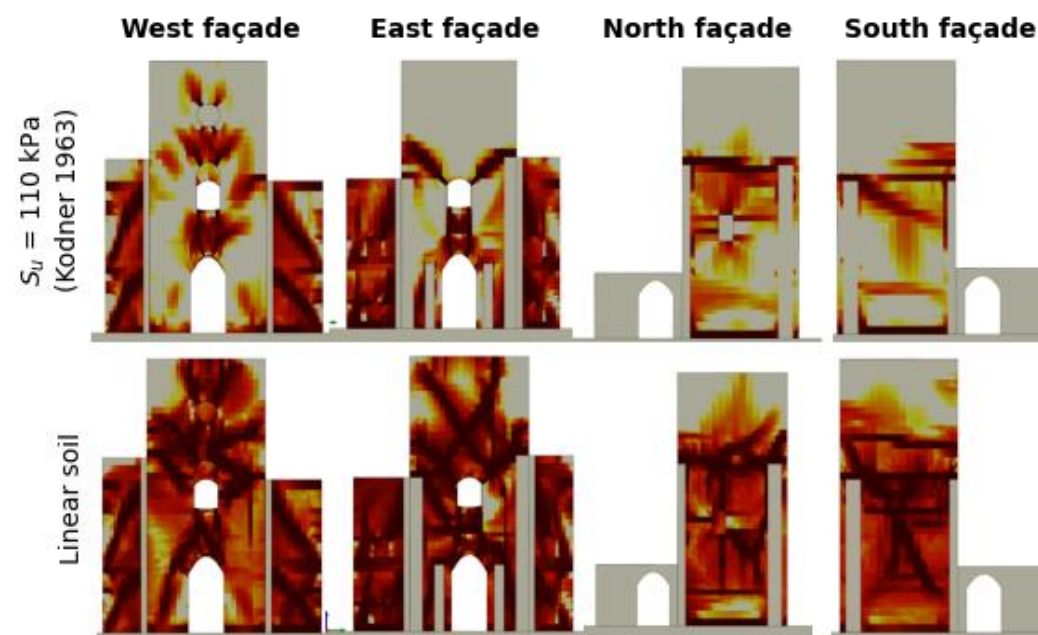
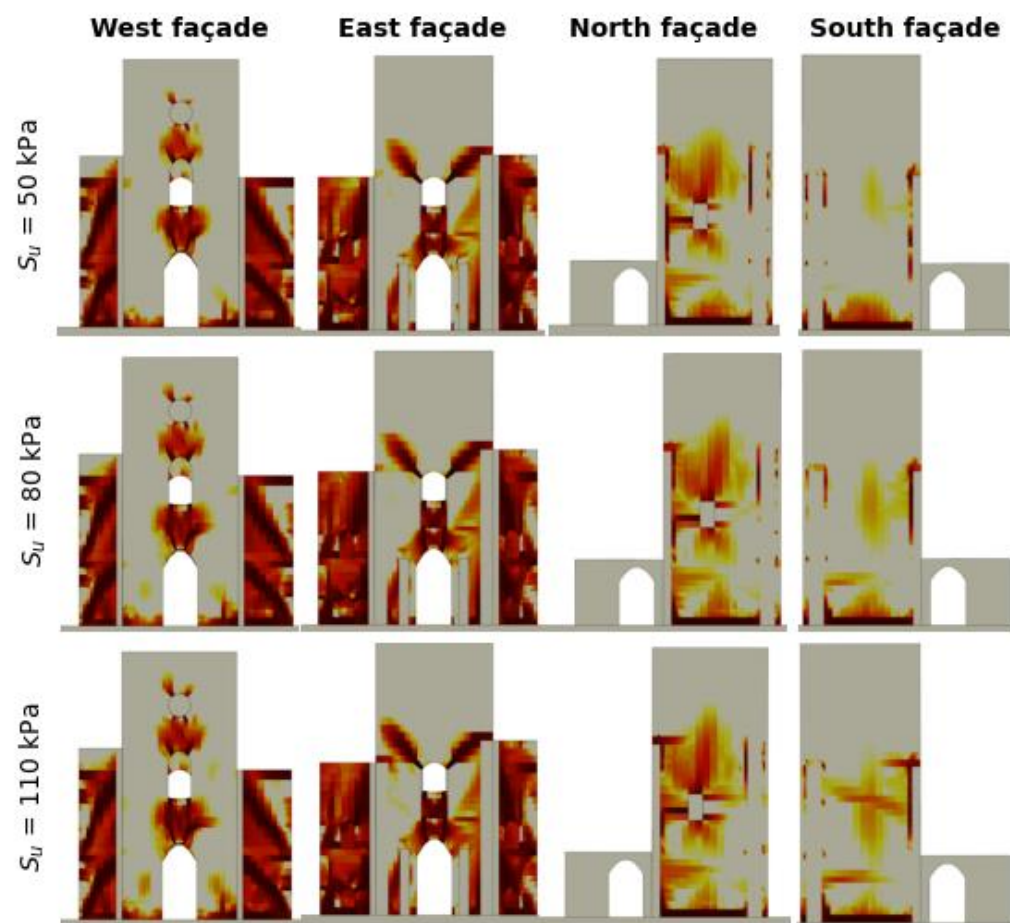
material damage (2 Components {d+, d-}; S)
d+



Linear soil

- Heavy damage like the no-soil fixed base case

Rocking isolation mechanism

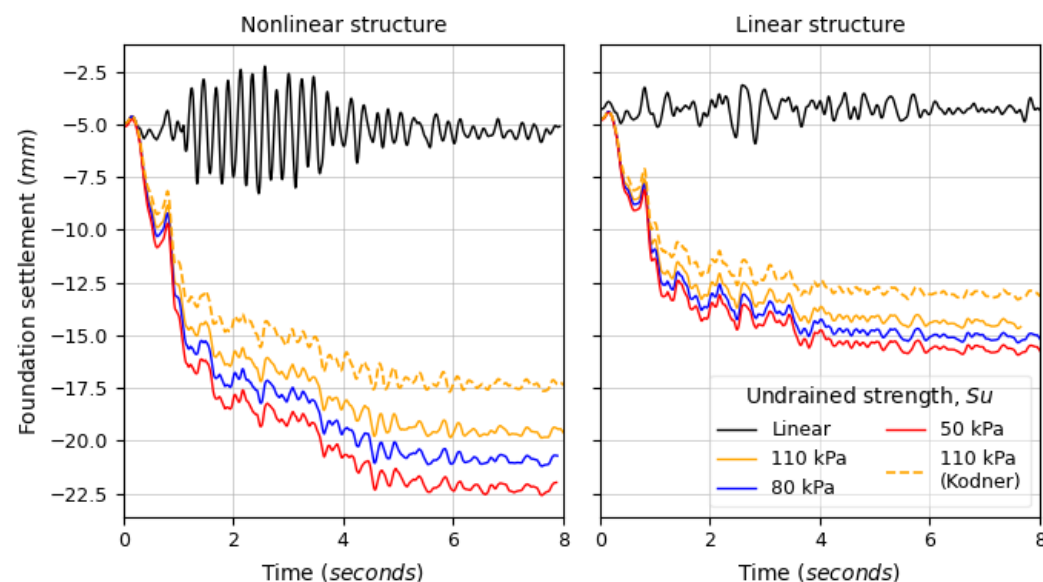


- It appears that G/G_{max} characteristics of the soil plays a greater role in the cracking damage response.

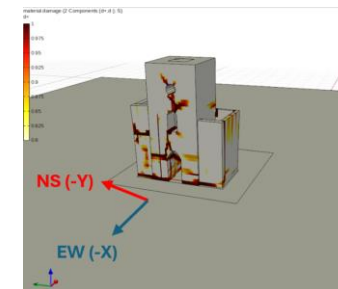
Rocking isolation mechanism

Tower is allowed to
soften

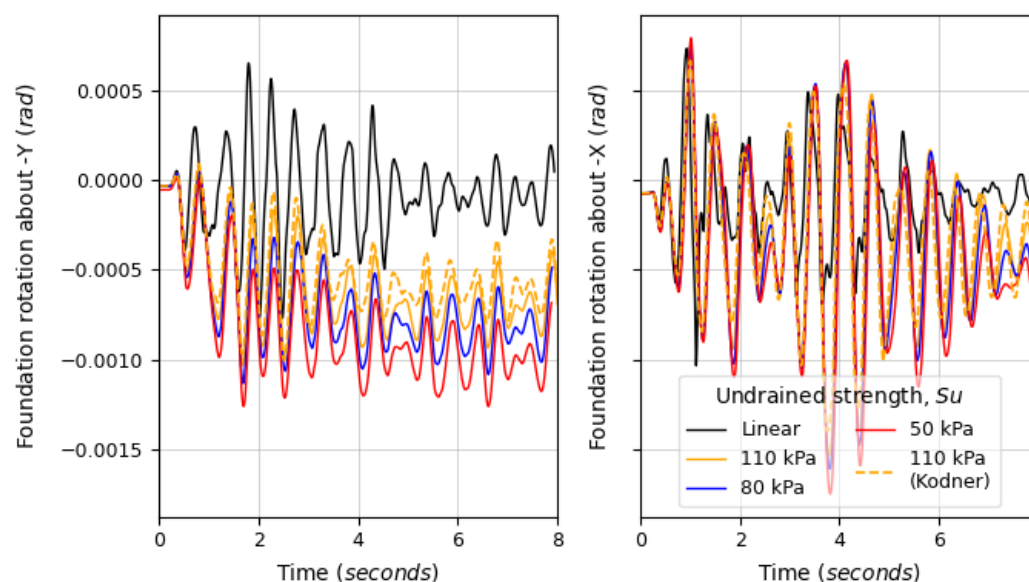
Tower is **not** allowed
to soften



EW



NS



1. *Trade off*: More displacement less seismic forces
2. Foundation softening leads to more settlements

Rotations are limited on the
contrary of what one expects!

Conclusions

- Some key take-aways:
 1. The **IMPL-EX integration** method can effectively address complex masonry-soil interaction problems, including contact nonlinearities like seismic pounding.
 2. Masonry towers exhibit a **rocking isolation mechanism**. As discussed by Gazetas (2015) for RC frames.
 3. In volume-conserving analyses, the soil's **stiffness degradation** significantly influences the extent of masonry damage.
- Some limitations:
 1. Soil behavior is modelled in **total stress conditions** which may model fine grained soil behavior. Shallow foundations behave differently under effective stresses as shown by Bullock et al. (2019), Macedo & Bray (2018).
 2. Important soil behavior features such as **strain softening** and **cyclic degradation** are not considered which could potentially limit the beneficial effect of the rocking isolation mechanism.

Thank you!

For videos and more, please visit the Research tab at:

www.onurdenizakan.com

in the following weeks (under development)

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Our Research Team



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Q&A

More slides on selected topics...

More on the IMPL-EX solution

1. Explicit phase

1. Extrapolate material state variables e.g., slip multiplier, λ :

$$\lambda_{n+1} = \lambda_n + (\lambda_n - \lambda_{n-1}) \cdot \text{time_factor}$$

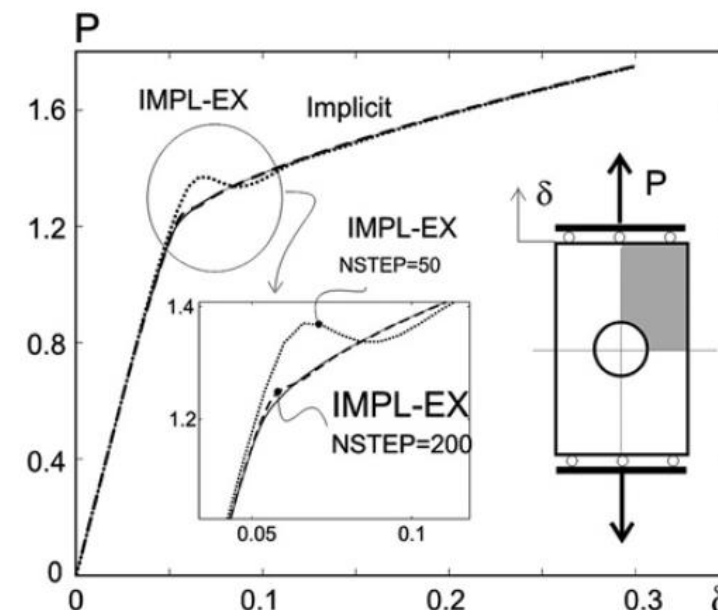
Additional state variable to store

2. Compute plastic strain increment, stress and tangent using the extrapolation (linear)

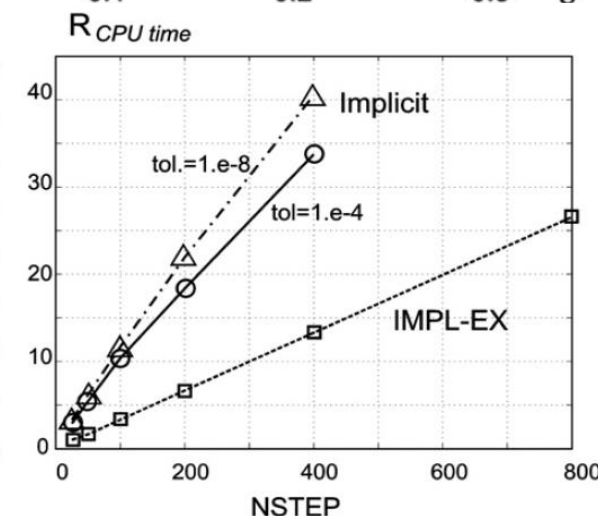
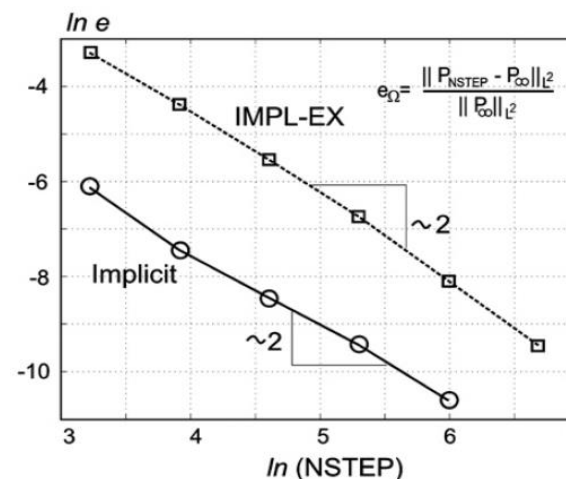
2. Implicit phase

1. **After Newton-Raphson** converges and **before the material commit** function is called
2. Re-compute state variables (λ_{n+1}) using implicit return mapping
3. Compute plastic strain increment and stress because of implicit solution
4. Store corrected state variables for extrapolation during next step

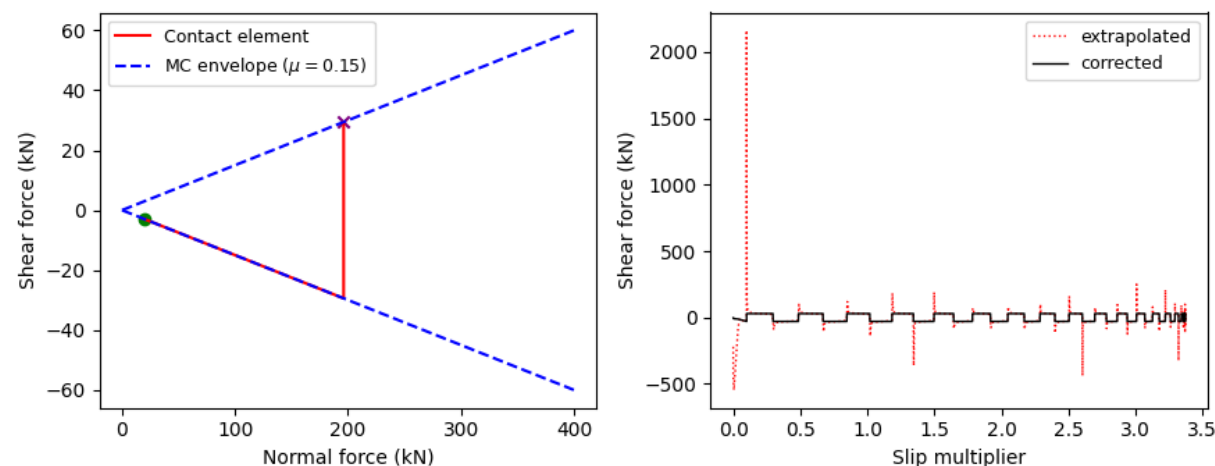
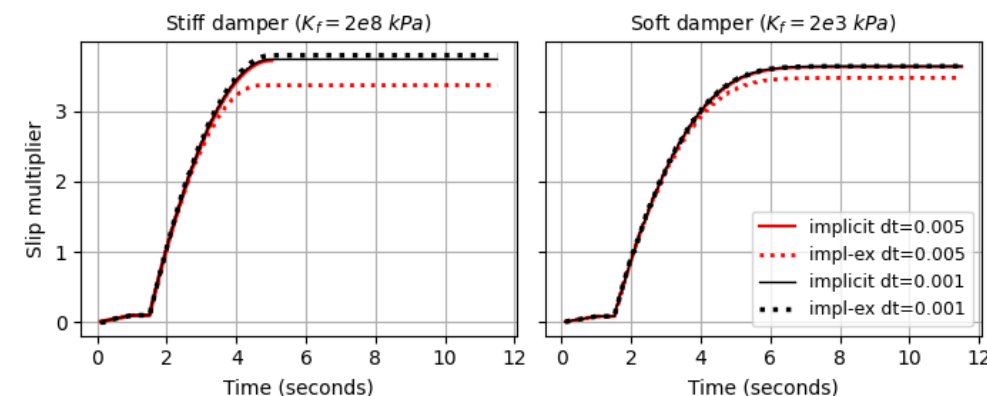
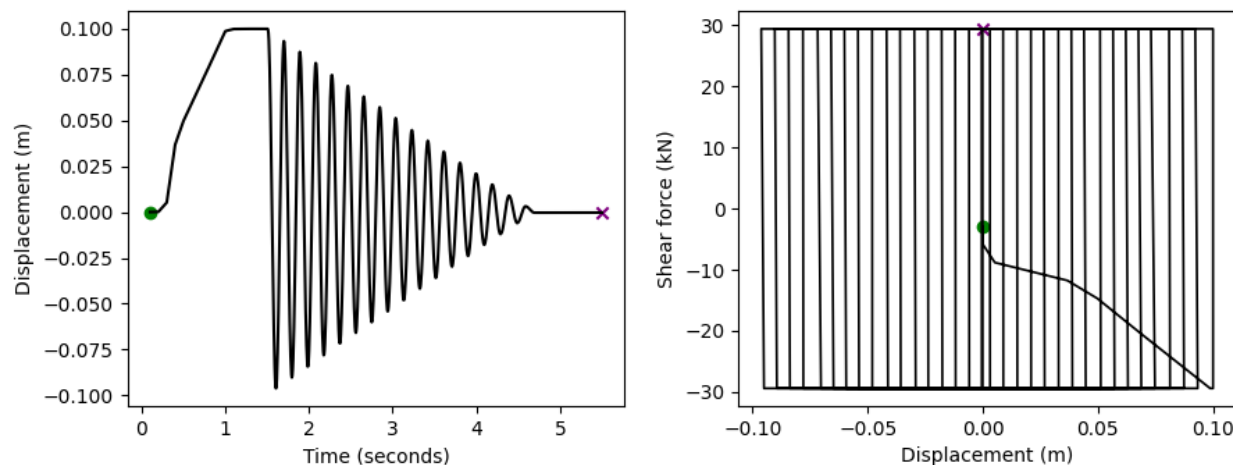
*Material equations need to be linearized for the explicit phase!



(Oliver et al., 2008)

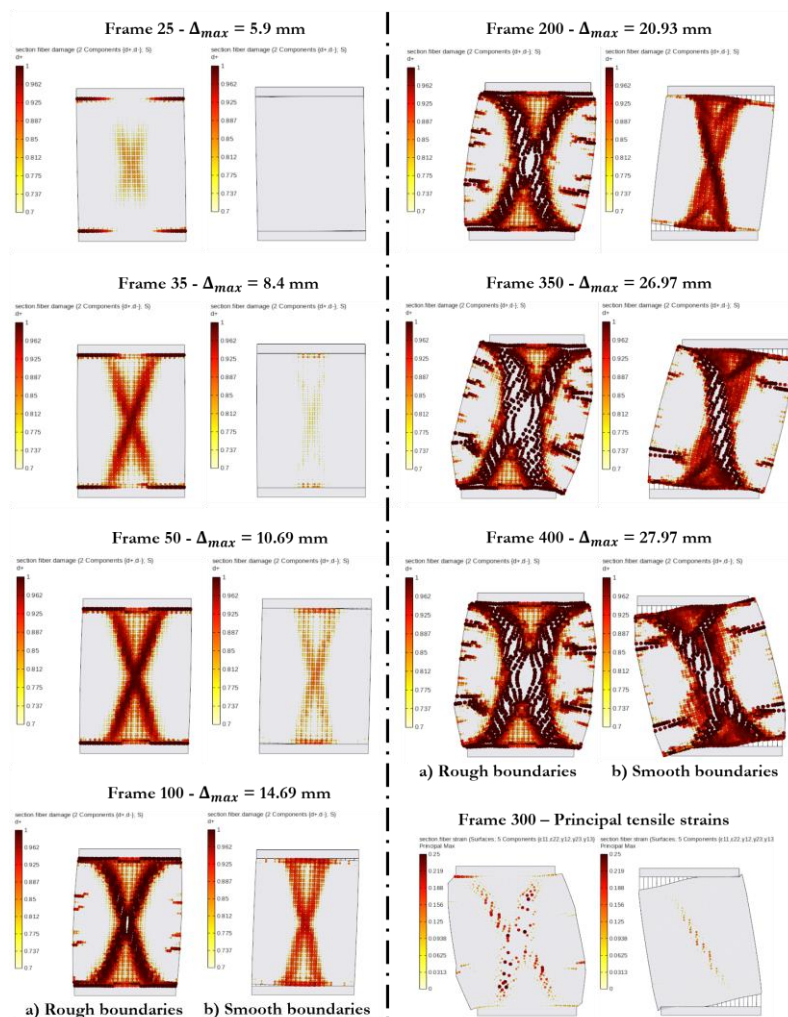


More on the contact element



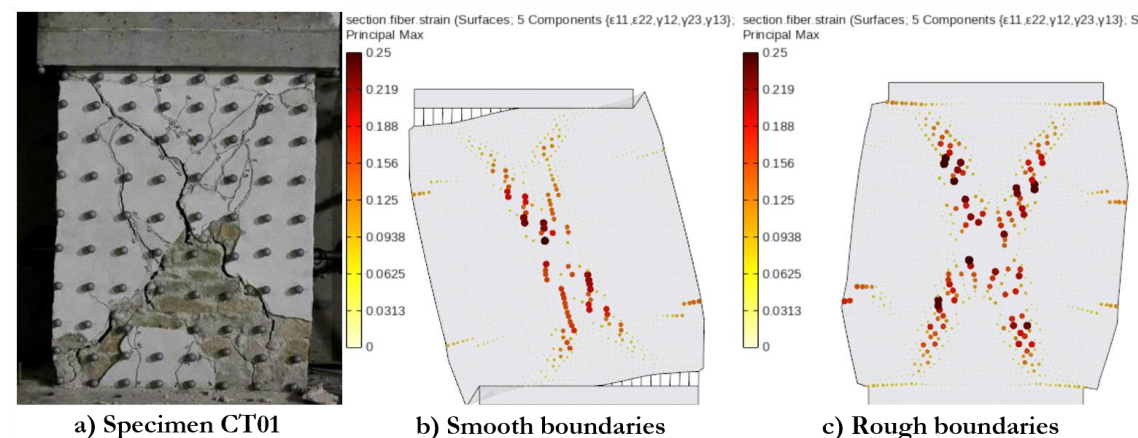
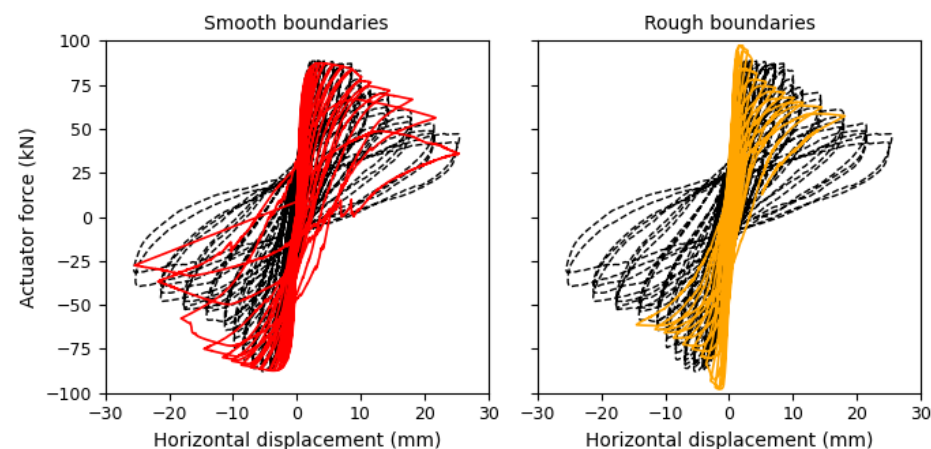
Soft impl-ex friction damper ($K_f = 2e8 \text{ kPa}$). The discrepancy between the extrapolated and corrected stresses. ($dt = 0.005 \text{ seconds}$)

More on the contact element

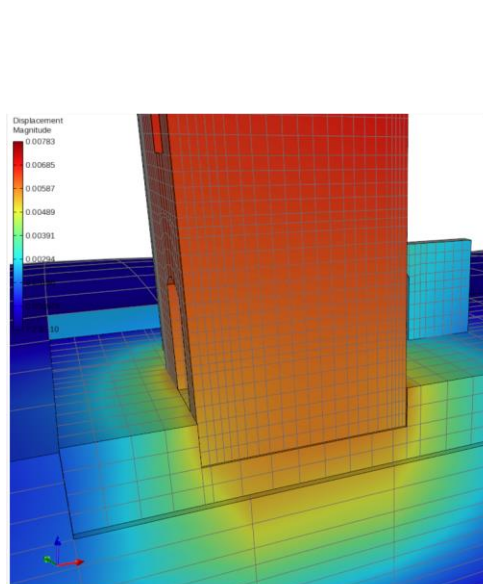


Comparison of masonry pier model:

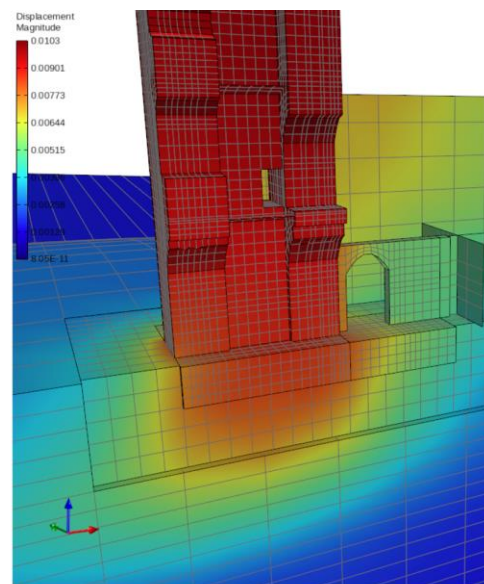
- a) with contact elements (smooth b.c.)
- b) without contact elements (rough b.c.) between rigid beam and the specimen.



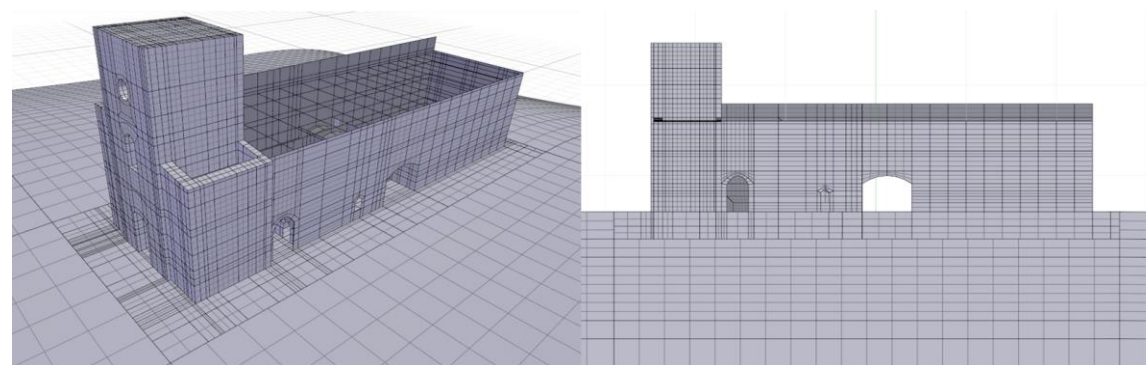
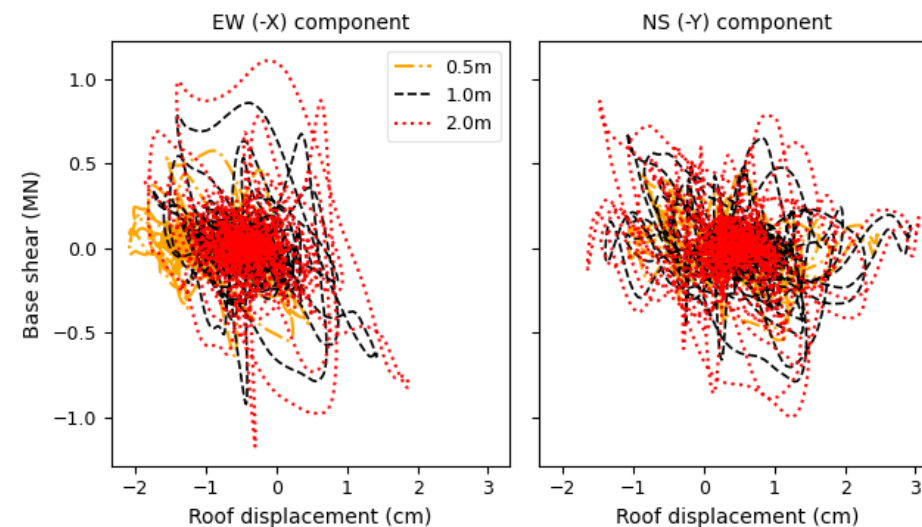
More on the transition soil model



a) Irregular transition soil mesh
(Abandoned)



b) Regular transition soil mesh
(1x1x1m elements)



The mesh dependency in the foundation response
A non-local multi-yield surface model??

References

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